MILITARY SPECIFICATION

CLOTH, TWILL, POLYESTER AND COTTON (WATER REPELLENT)

This specification is approved for use by Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of polyester and cotton twill cloth with a durable water repellent finish.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

MILITARY

MIL-T-43548 - Thread, Polyester, Cotton-Covered and Rayon-Covered

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to Federal Trade Commission, Washington, D.C. 20580.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted sahll be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) D 1424-63

Test for Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and uniformity of shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.3 and 6.2).

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3.3 Materials.

3.3.1 Fibers.

3.3.1.1 Polyester fibers. The fiber shall be polyethylene glycol terphthalate when tested as specified in 4.5.

3.3.1.2 Cotton fibers. The cotton shall be carded and combed when tested as specified in 4.5.

3.3.2 Yarns. The yarns shall be a blend of polyester and cotton, drawn and evenly spun into singles yarn for both warp and filling when tested as specified in 4.5.

3.3.3 Fiber content. The finished cloth shall contain 65 ± 5 percent polyester and 35 ± 5 percent cotton based on the dry desized weight of the cloth prior to the water-repellent treatment when tested as specified in 4.5.

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical requirements

Weight oz/sq. yd (g/m^2)	Yarns per inch (2.5 cm)		Breaking strength (lbs) (Newtons)		Tearing strength (lbs) (Newtons)	
Min.	Min.		Min.		Min.	
	Warp	Filling	Warp	Filling	Warp	Filling
7.8 (264.4)	110	48	240 (1068.0)	70 (311.5)	5.0 (22.3)	5.0 (22.3)

3.4.1 Weave. The weave shall be a $\frac{3}{1}$ left hand twill, when tested as specified in 4.5.

3.4.2 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum width, inclusive of selvages. When the cloth is manufactured with shuttleless loom selvages, the width of the cloth should be as specified (see 6.2) and shall be the minimum width of the woven cloth, excluding the selvages.

3.5 Color. Unless otherwise specified (see 6.2) the color of the finished cloth shall be shade Blue 3362 and shall match the standard sample.

3.5.1 Matching. The color of the finished cloth shall match the standard shade under artificial daylight, having a color temperature of 7500 ± 500 Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2850 ± 100 Kelvin.

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3.5.2 Colorfastness. The finished cloth shall show fastness to laundering (after 3 cycles), light (40 SFH), crocking, dry cleaning, wet-drycleaning, and perspiration equal to or better that the standard sample when tested as specified 4.5. When no standard sample is available, the finished cloth shall show a minimum of "good" fastness to laundering (3 cycles), perspiration, dry cleaning and wet-dry cleaning, a minimum of "fair" fastness to light (40 SFH), and a Munsell Value of not less than 8.5 for crocking when tested as specified in 4.5.

3.5.3 Sulfur content. The use of sulfur dyes and dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When no standard sample is available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.6 Finish. The cloth shall be scoured, dyed, heat set, and treated with an approved fluorocarbon type water repellent treatment.

3.6.1 Water repellency. The finished cloth shall conform to the requirements in Table II and 3.6.2 when tested as specified in 4.5.

Table II - Water repellency requirements

	Hydrostatic height min. centimeters	Dynamic absorption maximum
Initial	23	10%
After one dry cleaning	19	-
After three launderings	22	18%

3.6.2 Spray rating. The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than ratings 100, 100, 100 initial and 70, 70, 70 after one dry cleaning when tested as specified in 4.5.

3.7 Nonfibrous material. Prior to application of the water repellent treatment the starch and protein content (including chloroform-soluble and water-soluble material) shall not exceed 2.0 percent when tested as specified in 4.5.

3.8 Seam efficiency. The cloth shall have a seam efficiency of not less than 85.0 percent when tested as specified in 4.5.

3.9 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.10 Resistance to organic liquid. The finished cloth shall show no wetting by n-tetradecane either initially or after 3 launderings when tested as specified in 4.5.

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3.11 Dimensional stability. The finished cloth shall be preshrunk and shall not shrunk or elongate more than 3.0 percent in either the warp or filling direction when tested as specified in 4.5. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket or package.

3.12 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards (45.4 m). Each shall be put up on full width rolls as specified in PPP-P-1134.

3.13 Fiber identification. Each roll of cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act.

3.14 Marking. The face side of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.15 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (83.6 m^2), when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article of finished cloth shall be visually inspected for uniformity of shade, color, finish and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.4 Quality conformance inspection . Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or appliable purchase document.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of 4.4.2.1 through 4.4.2.4.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on face side only. When the total yardage in the roll does not exceed 100 yards (91.4 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4 m), only 100 yards (91.4 m) shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly visible at normal inspection distance (3 feet) (.9 m) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), E for slubs and D for knots, shall be scored. No linear yard (increments of 1 yard (.9 m) on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards (83.6 m^2) of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards (83.6 m^2) of two or more individual rolls exceed 38.0 points. If one roll exceeds the point level, a second sample of 20 rolls shall be examined only for individual roll quality examination. If one or more rolls exceed the individual point value, the lot shall be rejected. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards (83.6 } m^2\text{)}$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches (76.2 mm) or less in any dimension	- one point
For defects exceeding 3 inches (76.2 mm), but not exceeding 6 inches (152.4 mm) in any dimension	- two points
For defects exceeding 6 inches (152.4 mm), but not exceeding 9 inches (228.6 mm) in any dimensions	- three points
For defects exceeding 9 inches (228.6 mm) in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Overall uncleanness
Shade defects - shaded side to side, side to center, or end to end
Excessive neppiness

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4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards (1.82 m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.4.2.2.2 Total yardage in the sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.2.3 Examination for shade.

4.4.2.3.1 Individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.5.1). A roll shall be unacceptable if it fails to meet the requirements for shade match. The sample unit shall be a 4 by 20 inch (10.2 by 50.8 cm) swatch of the cloth. The sample unit shall be drawn from each roll in the lot.

4.4.2.4 Examination for face identification, identification of preshrinking process and non-conformance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends on the wrong side, two or more rolls in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking comply with the Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table III, shall be followed. The physical and chemical values specified in Section 3, except when otherwise indicated, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result.

The sample unti shall be a 1/2 yard (0.45 m) full width of the cloth prior to application of the water-repellent treatment for determinations of the non-fibrous materials content and fiber content, and 5 continuous yards (4.6 m) full width of the finished cloth for all other physical and chemical tests. The lot size shall be expressed in units of 1 yard (.9 m). The sample size shall be as follows and the lot shall be unacceptable if one or more sample units fail to meet the requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
802 up to and including 22,000	3
22,001 and over	5

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Table III - Test methods

Characteristic	Requirement reference	Test method
Yarn		
Polyester identification	3.3.1.1	1/
Cotton identification	3.3.1.2	1/
Cotton combed	3.3.1.2	1/
Yarn ply (singles)	3.3.2	Visual 2/
Fiber content		
Polyester (percent)	3.3.3	2100 4/5/1
Cotton (percent)	3.3.3	2100 4.5.1
Weight	3.4	5041
Yarns per inch (Warp and Filling)	3.4	5050
Breaking strength (Warp and Filling)	3.4	5100
Tearing strength (Warp and Filling)	3.4	ASTM D 1424-63
Weave	3.4.1	Visual 2/
Colorfastness to:		
Laundering (after 3 cycles)	3.5.2	5610 3/ 4/
Light (40 SFH)	3.5.2	5660
Crocking	3.5.2	5651
Perspiration	3.5.2	5680
Wet-dry cleaning	3.5.2	5622
Dry cleaning	3.5.2	5621
Sulfur content	3.5.3	2020
Finish	3.6	1/
Hydrostatic pressure		
Initial	3.6.1	5514
After 3 launderings	3.6.1	5556 and 5514 6/
After 1 drycleaning	3.6.1	5508 and 5514
Dynamic absorption		
Initial	3.6.1	5500
After 3 launderings	3.6.1	5556 and 5500 6/
Spray rating		
Initial	3.6.2	5526
After 1 dry cleaning	3.6.2	5508 and 5526
Nonfibrous materials	3.7	2611
Seam efficiency	3.8	5110 7/
pH	3.9	2811
Resistance to organic liquid		
Initial	3.10	4.4.2
After 3 launderings	3.10	5556 and 4.5.2 6/
Dimensional stability	3.11	5556 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

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3/ The specimen must be dried after each complete laundering cycle.

4/ Only the stain on the polyester and cotton fibers of the transfer cloth shall be evaluated.

5/ Cotton procedure shall be used.

6/ Specimens shall be subjected to 3 complete cycles (wash and dry) prior to determination of hydrostatic height, dynamic absorption, and resistance to organic liquid after laundering.

7/ Needle size 0.040 ± 0.001 inch (1.01 ± 0.02 mm) across the blade eye. The thread shall be polyester-covered in accordance with MIL-T-43548, Ticket No. 30, 3 ply for both the needle and bobbin. Seam type shall be LSq-2, stitch type 301, 8 to 10 stitches per inch, 3/8 inch (9.52 mm) gauge.

4.5.1 Fiber content. The general procedure of method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}} = \text{percent polyester}$$

100 - percent polyester = percent cotton

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent cotton of the two specimens shall be reported to the nearest 1.0 percent.

4.5.2 Test for resistance to organic liquid. Place a small specimen of the cloth on a smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After one minute, examine the specimen under light at an angle. Absence of light relfectance at the fabric-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall be considered cause for rejection of the lot represented by the sample.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

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5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The fabric covered by this specification is intended for use in raincoats worn by Coast Guard personnel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Color of cloth if other than specified (see 3.5).
c. Width required (see 3.8).
d. Minimum length if other than specified (see 3.12).
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When a first article inspection is required, the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, IL 69684.

Custodian:

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Review activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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